

GT-GÉOALGO: SCIENTIFIC TRENDS OVER THE LAST 20 YEARS

General Presentation

The field of research of the GT GéoAlgo is *Discrete and Computational Geometry*.

Computational Geometry studies computational questions related to geometric objects. These questions often come from applied fields such as robotics (motion planning), computer graphics (mesh processing), data mining (multidimensional search), or optimization (linear programming). The focus is on provably correct algorithms with worst-case theoretical guarantees. Some algorithms are numerical and imply precision issues, but in most cases one assumes exact computations on real numbers, and the algorithms have a discrete flavor. In the 2000s, *Computational Topology* emerged as a subfield of Computational Geometry and has grown significantly, in particular in the French community.

Discrete Geometry, also called *Combinatorial Geometry*, studies combinatorial aspects of geometry. It is naturally very interrelated with Computational Geometry, and should not be confused with *Digital Geometry*, which studies discretization of geometric objects on grid-like spaces (with pixels and voxels).

The flagship venue of the community is the *International Symposium on Computational Geometry* (SoCG). Additionally, publications occur in broader conferences in theoretical computer science and algorithms (STOC, FOCS, SODA), in machine learning (AISTATS, ICML, NeurIPS), and in more applied fields (Siggraph, SGP). Members participate in workshops in computer science (Dagstuhl) and mathematics (Oberwolfach). The community has dedicated journals (*Discrete & Computational Geometry* and *Journal of Computational Geometry*), but also publishes in journals with a broader scope (J. ACM, SIAM J. Comput., FOCM, and journals in algorithms, combinatorics, or geometry). The French community plays a vibrant role on the international scene. It gathers researchers from CNRS (sections 2 and 3, and to a lesser extent section 1), Inria, and many universities, and holds a biyearly meeting, the *Journées de Géométrie Algorithmique*, supported by the GDR. Below, we review the main trends of the last 20 years, with a strong focus on the topics that are studied by the French researchers.

1 Geometric algorithms in $\mathbb{R}^d$ with theoretical guarantees

Delaunay triangulations, Voronoï diagrams, and polytopes. These fundamental concepts are studied since the beginnings of Computational Geometry and they remain a vibrant area of research. Over the last twenty years, while new algorithms have been developed in the traditional Euclidean setting, the focus on Delaunay triangulations and Voronoï diagrams has also widely expanded: refined algorithms have been developed for weighted and constrained variants (see e.g. [22] for an application in computational optimal transport), for robustness in practical meshing pipelines, and for extensions to more complex metric spaces—including curved geometries such as hyperbolic surfaces (see below). At the same time, the study of convex polytopes has continued to grow, driven by the need to handle high-dimensional data and models. Because polytopes admit different representations whose sizes can significantly vary, designing efficient algorithms to approximate a polytope—or even estimate its volume—has become a key challenge. These lines of work illustrate how classical geometric structures continue to evolve in response to modern computational demands.

Geometric algorithms. Standard algorithmic problems are revisited in a geometric context. Prominently, any family of geometric objects can be studied through its *intersection graph*, and classical packing and covering problems (and variations) are attacked, sometimes in an approximate way, by exploiting properties of the intersection graph and by using other properties and concepts (VC-dimension, ε -nets, ...). For this purpose, algorithms on intersection graphs, e.g. for coloring, TSP, and separators, have been developed. As a particularly fruitful example, efficient approximation schemes have been discovered recently using local search, leading to algorithms that are very simple (but highly non-trivial to analyze). For the specific case of the plane, other types of graphs are relevant, such as *visibility graphs*, whose construction and properties have been widely studied until around 2015.

Numerical and symbolic computations. As mentioned above, most geometric algorithms are designed under the assumption that computations are performed over the real numbers. However, implementations do not manipulate real numbers but primarily fixed-precision floating-point numbers, possibly arbitrary-precision floating-point numbers or integers, and sometimes algebraic numbers. By the late 1990s, the community had converged on the consensus that implementations must return mathematically exact answers to geometric predicates (e.g., deciding whether a point lies to the left, to the right, or on a line defined by two other points). Without such guarantees, implementations can behave unpredictably crashing, entering infinite loops, or producing incorrect output. A notable example was identified in 2018: a [vulnerability in Skia graphics library](#) (used by Chrome etc.) caused by a non-state-of-the-art implementation of a 2D convex-hull algorithm. Answering predicates typically amounts to evaluating the sign of a polynomial at a given value, handled through numerical filtering: a first pass uses floating-point arithmetic together with error bounds, and exact computation is invoked only when this filter fails. These filters work very well for algorithms computing combinatorial structures, but they are insufficient for algorithms that generate new points or other geometric objects, a problem that has motivated several major research directions over the past two decades.

Key results include the first algorithm to remove intersections in 3D triangle meshes with fixed-precision output vertices while bounding the input-output distance [16, 32], addressing a long-standing practical need for industrial and academic implementations. Another result is the development of efficient algorithms for drawing algebraic plane curves, approximating both the curves and self-intersections while preserving topology and controlling approximation error [19]. A further achievement is the first practical and robust computation of 3D quadric intersections using algorithms that reduce the degree of algebraic numbers involved [6]. A recent trend focuses on hyperbolic geometry where cascaded constructions of points make it critical to design algorithms with predicates and algebraic numbers of degree as low as possible.

Probabilistic analysis of geometric algorithms. Over the past two decades, the probabilistic analysis of geometric algorithms has gained prominence as a way to temper the often pessimistic nature of worst-case bounds—such as the classical exponential upper bounds for the simplex method. This shift has motivated the introduction of more realistic input models and average-case analyses. A major conceptual breakthrough in the early 2000s was the development of smoothed analysis, which blends worst-case and average-case perspectives; its application to the simplex algorithm by Spielman and Teng had a profound impact on the field. In France, this line of research fostered closer interactions between communities (GTs) in analysis of algorithms and stochastic geometry, notably through several national collaborative projects in the 2010s. Over the last fifteen years, French researchers have made significant contributions in low-dimensional settings, including advances in smoothed analysis, probabilistic approximation of polytopes, and the study of random walks in geometric contexts. These works illustrate the increasingly rich interplay between geometry, randomness, and algorithmic efficiency.

2 Meshes, triangulations, geometric and topological inference

Mesher/triangulations and in particular *Delaunay complexes* have been a cornerstone of computational geometry from its inception. The interest in meshing is motivated by e.g. applications in numerical PDEs (finite element methods) and graphics. Up to the year 2000 the study of meshing was limited to low dimensions and mostly Euclidean space. Since then, the research has gone into new directions with an emphasis on high dimensions and curved spaces: surfaces, hyperbolic spaces, and more generally Riemannian manifolds. The manifolds we want to mesh occur in various forms: submanifolds that are known from samples (this combines meshing with shape reconstruction discussed below), submanifolds that are given by (not necessarily algebraic) equations, or abstract Riemannian manifolds (given by local charts ignoring the ambient space, which avoids the curse of dimensionality). For instance, one important line of work has been to adapt the classical algorithmic toolbox for Delaunay triangulation so that it works beyond Euclidean settings. This has been initiated in toroidal and hyperbolic geometries, both of which are relevant for applications, e.g., in materials sciences, and many questions still abound in this active area of research. The works in the community have many facets: algorithm development, theoretical guarantees of correctness, complexity theory, implementation and practical running time.

Shape reconstruction is a classical problem in computational geometry, closely related to meshing. Given a finite set of data points that sample an unknown shape, the problem consists in building an approximation (ideally a triangulation) of the shape based solely on the data points. In the early 2000s, the first methods with theoretical guarantees were proposed for surfaces in $\mathbb{R}^3$. Around 2005, the focus started to shift toward *manifolds* in high dimensional ambient spaces, motivated by applications to *machine learning*. A milestone was obtained with the work of Niyogi, Smale and Weinberger in 2008, which provided sampling conditions under which an offset of the

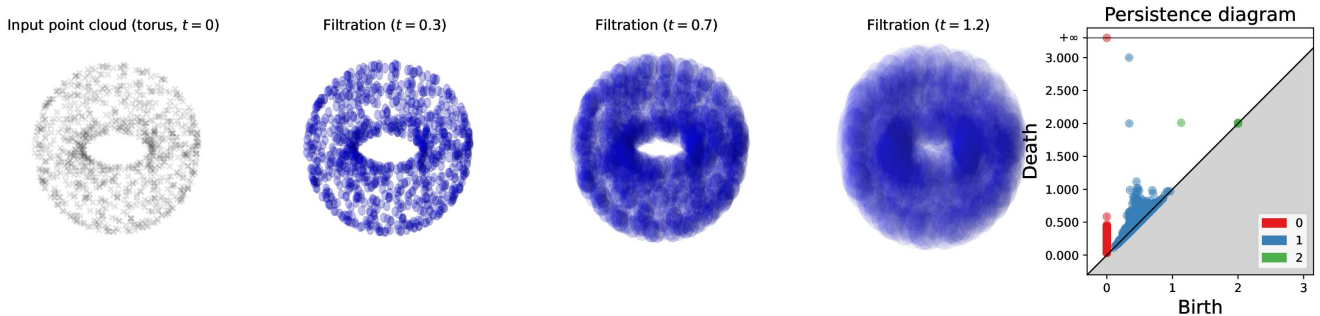


Figure 1: Čech filtration and resulting persistence diagram built on top of a point cloud (1000 points sampled uniformly on the surface of a torus). The right plot depicts the resulting PD: red points correspond to homology of dimension 0 (connected components), blue points to dimension 1 (loops) and green points to dimension 2 (cavities). The two prominent blue points and the green one testify for the two loops generating the torus and the corresponding cavity, points closer to the diagonal (birth $\simeq$ death) correspond to features that are merged quickly after appearing in the filtration (i.e. persisted less longer), often considered as topological noise.

data points recovers the homotopy type of the underlying manifold [31]. Since then, significant effort of the French community has gone into generalizing and improving the bounds of that seminal paper (e.g. [9, 2]). In a high-dimensional ambient space, classical algorithms based on the computation of the Delaunay complex of the data points suffer from the curse of dimensionality. Alternative methods have been proposed that scale well with the ambient dimension and come with guarantees [4]. This led to developing new data structures for encoding simplicial complexes: the simplex tree and the blocker-skeleton data structure, which are both at the core of the Gudhi library (cf. §3 and 6). In a recent breakthrough, manifold reconstruction has been expressed as finding minimal cycles (either for an ℓ_1 -norm [3] or for a lexicographic order, [12]), yielding to a topologically correct and efficient algorithm for surfaces in $\mathbb{R}^3$. [11]. The geometric assumptions that underpins the correctness of many algorithms mentioned above are most often formulated in terms of the reach [20]: the largest distance under which a point is guaranteed to admit a unique projection on the manifold. A new metric characterization of the reach was provided in [5] which proved useful for (statistical) learning. The reach is intricately linked with the medial axis, a skeleton which lies in the middle of a shape and captures the homotopy type [27] and plays an important role in graphics, though being difficult to compute. Significant effort has gone into computing and simplifying the medial axis.

3 Topological Data Analysis

Topological Data Analysis is a relatively recent machinery in computational topology that aims at computing *quantitative* topological descriptors of structured objects (graphs, points sampled on a manifold, time series, etc.). The field emerged in the early 2000s, and has since found application domains such as computer graphics, material science and computational biology to name a few. Rooted in algebraic topology, and more precisely in the theory of *persistent homology*, the TDA pipeline can be summarized in the following way: given a topological space $\mathcal{X}$ and a map $f : \mathcal{X} \rightarrow \mathbb{R}$, consider the *filtration* $K_t = \{x \in \mathcal{X}, f(x) \leq t\}$. For instance, one may take $\mathcal{X} = \mathbb{R}^d$ and $f(x) = \min_{1 \leq i \leq n} \|x - x_i\|$ where $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^d$ is a point cloud, yielding so-called the Čech filtration. One then records the pairs of values (t_b, t_d) at which a given topological feature (such as connected components, loops, cavities, etc.) appears in K_{t_b} and disappears in K_{t_d} .¹ The resulting (multi-)set of pairs is called the *persistence diagram* (PD) $\text{PD}(f)$ of f . See Figure 1. A fundamental property of PDs is their *stability*: similar filtrations yield similar PDs [10], where similarity of PDs is measured using optimal transport-type distances [18].

By the mid-2010s, PDs began to be used in machine learning as static features derived from data. Because the space of PDs has a non-linear structure, their integration into downstream models is non-trivial, and typically relies on vectorizations techniques, such as kernel methods, or data-driven representations based on specific deep neural network architectures. More recently, to further strengthen the connections between TDA and machine learning, the concept of persistence-based *topological optimization* has emerged. A rigorous framework for differentiating composite maps of the form $f \mapsto \text{PD}(f) \mapsto \ell(\text{PD}(f)) \in \mathbb{R}$ for some loss functions ℓ as been introduced in [26].

¹In practice, K_t is represented by a combinatorial object called a simplicial complex (which should have, or closely approximate, the same homotopy type as the corresponding sublevel set), and PDs are computed using matrix reduction algorithms.

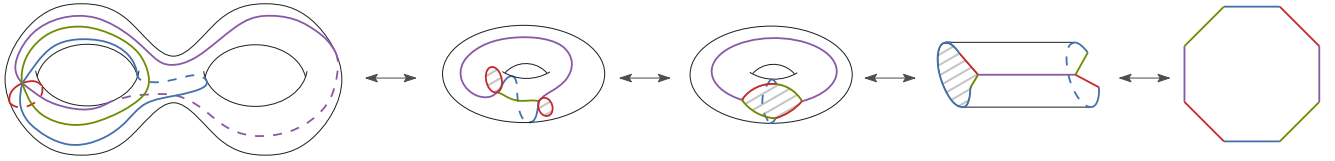


Figure 2: Cutting a double-torus (an orientable surface of genus two) into a disk using a *canonical polygonal schema*.

A series of works has since been devoted to studying, implementing, and refining gradient-based optimization schemes—an area that continues to be particularly active within the French TDA community.

In addition, the field has seen significant advances in the context of multiparameter persistence, that is, when the map f is valued in $\mathbb{R}^d$ with $d \geq 2$. In contrast to the case $d = 1$, the lack of total order on $\mathbb{R}^d$ prevents one to define canonical analogues of PDs. The design of topological features based on multiparameter persistence has been studied deeply in the field, with the French community being a driving force, with the design, theoretical analysis, and implementation of two topological features for multiparameter persistence: the *signed barcode* and *candidate decompositions*. Multiparameter topological optimization has then quickly followed as a new application field in TDA, fueled by these new features.

These efforts materialize by the development of the open-source library [Gudhi](#) [29] (Python / C++), which contains most of TDA descriptors with fast implementations in C++, as well as easy-to-use pipelines in Python for data science. As such, it has become the leading library for TDA, with thousands of downloads per week. With the same ambition, the development of multiparameter persistence has been made easier thanks to the [multipers](#) library, whose core implementations rely on [Gudhi](#).

4 Low-dimensional computational topology

While algorithmic aspects are implicit in topology since its inception, a systematic development of algorithms for topological problems emerged only in the late 1990s [17]. The French community is particularly active in low-dimensional computational topology, where the ambient space is a surface or a 3-manifold (a space locally homeomorphic to $\mathbb{R}^3$), with some non-trivial topology (e.g., “surfaces with handles” such as the torus). Like topological data analysis, this field relies on algebraic topology, but it has a very distinct flavor: the applications are very different, and low dimension allows us to work with homotopy, which is finer than homology. The relevant tools from computer science include graph algorithms, complexity theory, and fixed-parameter tractability. In this area, the GT-GéoAlgo has connections with several other GTs, notably CoA and Graphs, and is also related to other fields, such as graph drawing. Relevant applications are found in computer graphics, geometry processing, and molecular biology.

The **2-dimensional case**, namely computational topology for graphs on surfaces [13], started to flourish 20 years ago [25]. The most basic questions are algorithmic formulations of well-studied mathematical problems: Are two input curves or graphs on a surface equivalent, for various topological notions of equivalence (homotopy, homology, isotopy)? Given a self-intersecting graph or multicurve, what is the minimum number of crossings that must remain after homotopically deforming it on a surface? Around the same time, algorithms for computing shortest curves and graphs with given topological properties blossomed, e.g., shortest non-contractible or non-separating closed curves; this requires an appropriate discretization of a metric surface.

While the above line of research is still active, several directions have emerged more recently. Applications to graph problems, e.g., generalized cut and flow problems, have been found: combining topological subroutines with tools from algorithmic graph theory (treewidth, minors, etc.), faster algorithms are obtained assuming the input graph is planar or, more generally, is embedded on a fixed surface—often these algorithms are fixed-parameter tractable in the genus of the surface. Generalizations of planar structures (e.g., Schnyder woods) to surfaces are investigated. Compact data structures for surface triangulations are discovered. Efficient algorithms are described to compute well-studied mathematical quantities related to surfaces, such as their spectra. Algorithms and data structures are being developed to handle hyperbolic surfaces and compute Delaunay triangulations and Voronoi diagrams on such spaces.

In the **3-dimensional case**, new topological features and knotting phenomena start to appear. Their investigation from a computational point of view was initiated about twenty years ago in a seminal work [21] and our understanding has been steadily improving since then, see for example [24]. In terms of exact recognition algorithms, this area is populated with peculiar questions at all ends of the complexity spectrum. For instance, the problem of

deciding if a closed curve is knotted is now known to be in $\text{NP} \cap \text{co-NP}$ but no polynomial-time algorithm is known, while the best-known algorithm to check whether two knots are equivalent is elementary recursive but the problem is not even known to be NP-hard. Topological questions in 2D on homotopy, homology, etc. also adapt naturally to 3-manifolds, leading to new problems which are often harder to solve algorithmically. Over the past decades, the computational geometry community in France and abroad has largely contributed to mapping the 3-dimensional complexity landscape, in particular through the design of hardness proofs [15]. Another computational aspect of this field is that topological insights are often garnered via the development of highly-optimized specific software tailored for the analysis of knots and 3-manifolds [7, 14], which then fuels experimental works [8, 30].

Given the dearth of knowledge on exact recognition algorithms, it is natural to investigate *invariants* which provide a coarser way to differentiate topological objects. Topological invariants of algebraic nature, such as quantum invariants of knots and 3-manifolds whose introduction dates back to the 80s, have been recently studied from the computational point of view. Their computational hardness has been established thanks to connections with quantum computing [23], and efficient algorithms have been designed through the extension of techniques from parameterized complexity. This phenomenon has, in turn, fueled a new interest for the study of the structural properties of combinatorial representations of knots, 3-manifolds, and other low dimensional objects, notably with the extension of techniques from structural graph theory to the context of low dimensional topology [28], or the study of the complexity of hard problems on inputs of restricted topology.

5 Discrete Geometry, geometric and topological combinatorics

Discrete geometry studies the structure of discrete objects in a geometric space, from finite point sets in the plane to arrangements of n -dimensional convex bodies. Foundational problems in this area deal with packing (Kepler’s conjecture), polytopes and triangulations (Hilbert’s third problem on decomposing polyhedra, Hirsch’s conjecture on the diameter of polytopes, the g -conjecture on the face numbers of triangulated spheres), point configurations (the Erdős-Szekeres problem, distinct and repeated distances, integer distance sets), etc. There are strong ties between discrete geometry and computational geometry around questions of complexity, simulation, discretization, etc. and the two communities merged to some extent in the 1980’s. Within the French community, several lines of research in discrete geometry have been particularly active over the last two decades. Disk and ball packings in dimensions two and three continue to attract attention: although the monumental proof of the Kepler conjecture settles the case of a single radius, the landscape becomes far richer when balls with multiple radii are allowed, raising challenging questions of optimal structure and computational optimization. Progress has also been made on the study of oriented matroids, a combinatorial and algebraic structure arising from point configurations and hyperplane arrangements, with connections to the complexity class ETR (Existential Theory of the Reals), to probabilistic properties of geometric structures, and to refined combinatorial classifications. Hitting sets problems (finding a small number of points/lines/hyperplanes that intersect all elements of a collection of sets) have likewise seen strong advances, especially regarding the existence and computation of hitting sets and the influence that the geometry (convexity, half-spaces, etc.) has on the size of hitting sets. Equipartition problems (seeking for a single hyperplane cutting a collection of objects in halves, e.g. ham-sandwich theorem)—often involving partitions by systems of hyperplanes—have developed new links to classical conjectures such as Mahler’s. Beyond these themes, new combinatorial objects inspired by geometric constructions have been introduced, including generalized associahedra and related polyhedral families, and have deepened the study of reconfiguration graphs such as flip graphs of triangulations and decomposition complexes, often in connection with low-dimensional topology (section 4).

In the last 20 years, the development of algebraic and topological methods have been two driving forces in discrete geometry. A spectacular example was the *polynomial method*, which builds on ruled surface theory in real algebraic geometry and led to solutions to the *joint problem* and Erdős’ *distinct distances problem* and the algorithmic technique of *polynomial partitioning*. Another landmark is Adiprasito’s proof of the *g -conjecture* [1], which draws on the connections between face rings and toric varieties and yields extensions to simplicial complexes of some classical results on graphs (e.g. a linear bound on the number of facets in terms of the number of ridges, a generalization of Heawood’s inequality on which complexes embed in manifolds, etc.). Other examples include the sharpening of several fundamental results in extremal combinatorics for *Semi-algebraic (hyper)graphs*, the developments around the *configuration space/test map* scheme (e.g. the topological Tverberg theorem), etc.

6 Software development in Computational Geometry

The French community is involved in the development of several open-source software related to computational geometry: [gudhi](#) for Topological Data Analysis, contributions to **Maple** (root finding algorithms), [Isotop](#) (drawing of planar curves with correct topology), to name a few.

A major contribution of the French community to discrete and computational Geometry over the past decades lies in the development and long-term maintenance of the [Computational Geometry Algorithms Library \(CGAL\)](#). Initiated in 1996, CGAL has grown into the most influential and widely used software library in computational geometry. It provides robust, efficient, and well-documented implementations of geometric algorithms for both academic and industrial use. The project now comprises more than 700,000 lines of code, is downloaded about 10,000 times per year, and serves over 200 commercial users worldwide. With around 20 active developers and supervised by an editorial board in charge of project management and code review, CGAL continues to be a vibrant and sustainable initiative, supported by academic partners who have made long-term commitments to its maintenance and evolution. The library is distributed under both open-source and commercial licenses, ensuring accessibility for research as well as reliability for industrial applications. The private company GeometryFactory, an integral partner of the project (including 3 members of the editorial board), provides professional support and assists industrial users in integrating CGAL components into their software. In recognition of its enduring impact, CGAL received the “[Test of Time Award](#)” at the 2023 Symposium on Computational Geometry (SoCG), underscoring its role as a cornerstone in the field’s computational infrastructure.

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